# GOVERNMENT MODEL ENGINEERING COLLEGE, KOCHI

Department of Electronics and Communication Engineering
S7 Project Phase 1 - ECD415

## Field Programmable Gate Array Implementation of True Random Number Generation for On-chip Hardware Encryption

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|---|---|
| Project Guide | |
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Submitted by Group 4


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## ABSTRACT

Modern digital systems require secure, unpredictable random numbers for encryption, authentication, and secure communication. Software-based random number generators often lack sufficient entropy and consume significant processing resources. This project presents an adaptive FPGA-based True Random Number Generator (TRNG) integrated with an AES-128 encryption core, enabling direct generation of cryptographic keys from hardware physical characteristics.

The system architecture uses an array of ring oscillators—odd-numbered NOT gates in feedback loops—whose frequency variations and timing jitter provide the entropy source. Multiple oscillator outputs are combined via XOR and sampled with a system clock to produce raw random bits. A conditioning stage employing Von Neumann correction or cryptographic extraction improves bit quality before 128-bit AES key generation.

The core innovation is an adaptive control mechanism that monitors entropy quality through zero-one balance, pattern detection, run-length analysis, and oscillator frequency variation. When entropy degrades due to environmental changes, the controller dynamically responds by selecting alternate oscillators, disabling weak elements, adjusting sampling, or regenerating sequences—ensuring reliability beyond static TRNG architectures.

Implementation uses Verilog HDL on Xilinx FPGA, with modular design for ring oscillators, entropy extraction, conditioning, health monitoring, adaptive control, key generation, and AES-128 encryption. Each module undergoes simulation and testbench verification before synthesis and deployment. Performance evaluation includes randomness testing, quality metrics, operating frequency, throughput, resource utilization, and power consumption.

The system provides a compact, practical hardware security solution for IoT devices, embedded systems, wireless communication, access control, industrial controllers, smart cards, and security modules requiring secure key generation and on-device encryption.

**HARDWARE AND SOFTWARE REQUIREMENTS**

| No | Requirement | **Cost** |
|---|---|---|
| 1. | Artix-7 FPGA Development Board | 15000/- |
| 2. | UART Interface | 1000/- |
| 3. | Xilinx Vivado | |

**WORK DIVISION**

**Aditya S** : Ring Oscillators + Entropy Extraction

Generate randomness from silicon
- Design 3-5 ring oscillators (LUT-based loops)
- Implement phase sampling logic (capture oscillator outputs simultaneously)
- XOR outputs to extract random bits
- Verify jitter characteristics via simulation
- Optimize oscillator delays (use carry chains or LUT cascades)

**Anagha V Nair** : Key Accumulation + System Integration

Collect random bits into 128-bit key
- 128-bit shift register (accumulates extracted bits)
- Counter (tracks when 128 bits collected)
- key_ready signal generation
- Top-level instantiation of all modules
- Timing/clock domain management

**Harshadithyan V P** : AES-128 + Cryptographic Integration

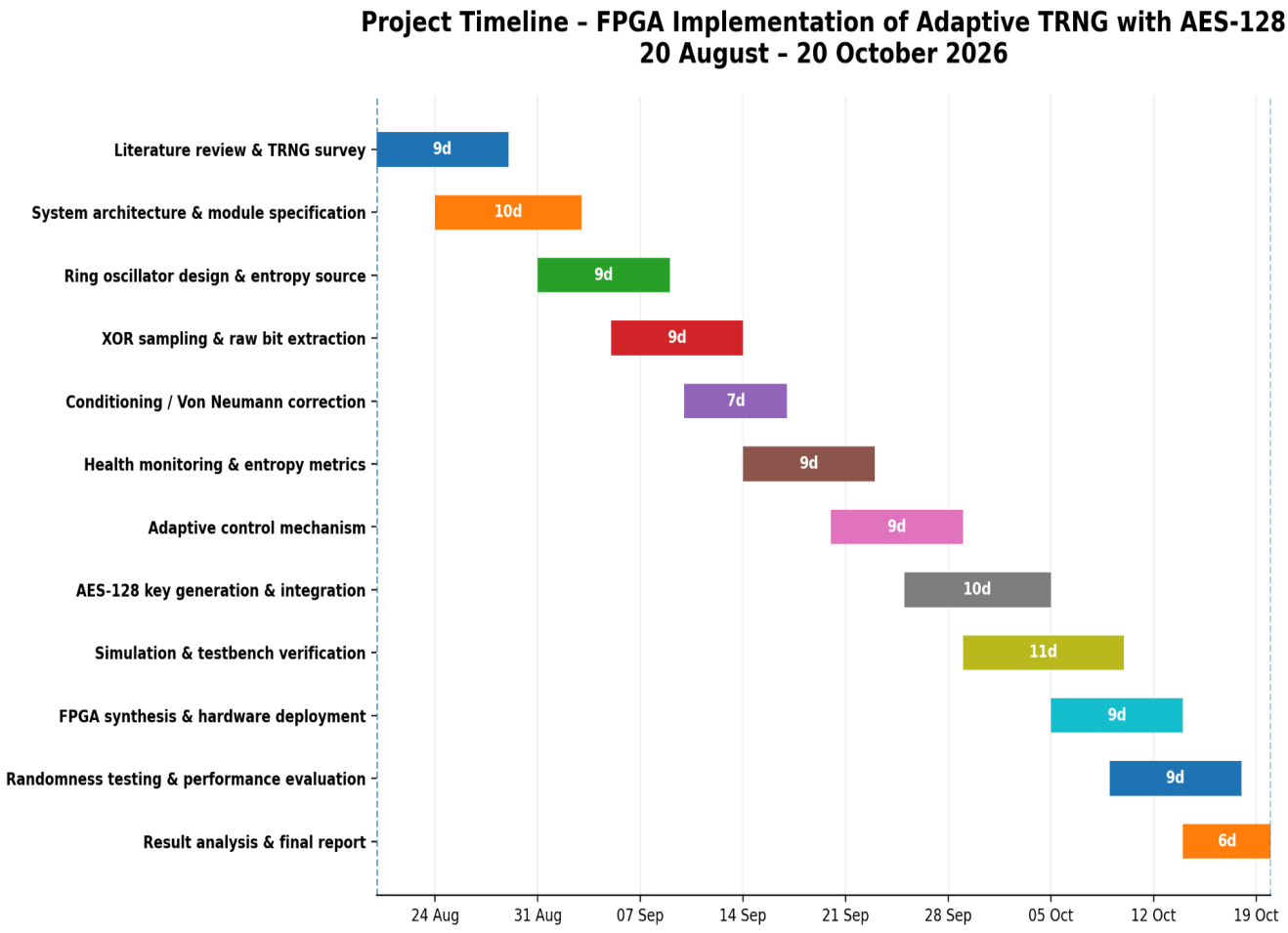
Encrypt with generated key
- Download open-source AES-128 core (secworks/aes from GitHub)
- Integrate AES with key from Person 2
- Hardcode plaintext: "HELLO_TRNG_DEMO!" (128-bit)
- Orchestrate handshake (wait for key_ready, trigger AES, capture output)
- Simulate full crypto pipeline

**Rufus Binoy Mathew** : UART + PC Verification

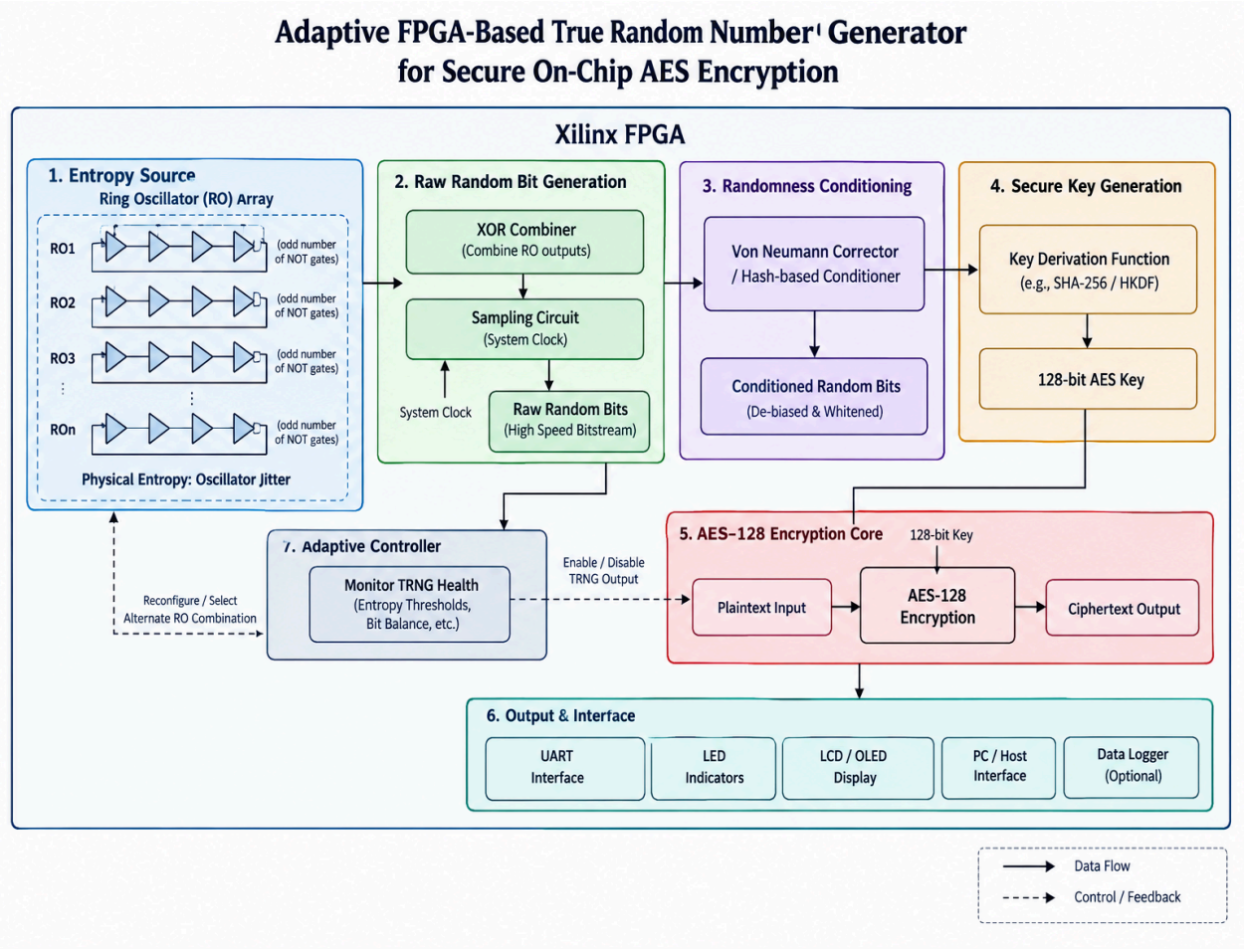
Send to PC and decrypt
- UART transmitter (115200 baud, standard)
- Format output: Key (32 hex chars) + newline + Ciphertext (32 hex chars) + newline
- Trigger on AES done signal

- Simple serial module (no handshake)

**GANTT CHART**



# SYSTEM ARCHITECTURE



## REFERENCE(S)

[1] M. A. Prada-Delgado, C. Martínez-Gómez, and I. Baturone, “Auto-Calibrated Ring Oscillator TRNG Based on Jitter Accumulation,” in *Proc. IEEE Int. Symp. Circuits Syst. (ISCAS)*, Seville, Spain, 2020, pp. 1–4, doi: 10.1109/ISCAS45731.2020.9180598.

[2] M. A. Şarkışla and S. Ergün, “Ring Oscillator Based Random Number Generator Using Wake-up and Shut-down Uncertainties,” in 2018 IEEE International Conference on..., 2018. [IEEE Xplore]

[3] S. Choi, Y. Shin, and H. Yoo, “Analysis of Ring-Oscillator-based True Random Number Generator on FPGAs,” in 2021 International Conference on Electronics, Information and Communication (ICEIC), 2021, pp. 1–4, doi: 10.1109/ICEIC51217.2021.9369714